Lecture: Stochastic reaction-diffusion systems

Reaction-diffusion systems are coupled systems of semilinear pde's including Laplace operators $\Delta = \sum_{i=1}^{d} \partial^2_{x_i x_i}$ (or more general strictly elliptic second order partial differential operators $\sum_{i,j=1}^{d} a_{ij} \partial^2_{x_i x_j}$) modelling diffusion and zero order terms of the type $f(v(t,x))$ modelling (chemical) reactions.

We will restrict to the scalar-valued case

$$(1) \qquad \partial_t v(t,x) = \nu \partial^2_{xx} v(t,x) + b f(v(t,x))$$

where $x \in D \subseteq \mathbb{R}^d$, $f: \mathbb{R} \to \mathbb{R}$, $\nu, b > 0$

Typically one assumes:

- (coercivity) $\lim_{|v| \to \infty} f(v) \cdot v = -\infty$
- (one-sided) Lipschitz condition $(f(v_1) - f(v_2))(v_1 - v_2) \leq L \|v_1 - v_2\|^2$.

Special cases (in $d=1$)

- $f(v) = v(1-v)(v-a)$ $\quad a \in (0,1)$ $\quad$ Nagumo equ
- $f(v) = v(1-v)$ $\quad$ Fisher resp. Kolmogorov-Petrovsky-Piskounov equ.

Travelling waves

are particular solutions to (1) of the type

(2) $v(t,x) = \hat{v}(x \cdot n + ct)$ $\quad$ c = velocity of the wave
$n \in \mathbb{R}^d$, $|n| = 1$, direction of motion of the wave

Inserting the Ansatz $v(t,x)$ into (1) yields

$$+c\hat{v}'(x \cdot n + ct) = \nu \hat{v}''(x \cdot n + ct) + b f(\hat{v})(x \cdot n + ct)$$

i.e., an ordinary differential equation in $\hat{v}$ of second order.

Examples
(Nagumo equation)

(3) $\quad + c\hat{v}' = \nu \hat{v}'' + b\,\hat{v}(1-\hat{v})(\hat{v}-a)$

Ansatz for explicit solution: $\hat{v}(x) = \dfrac{1}{1+e^{-kx}}$, $k \in \mathbb{R}$,
yields

$$\hat{v}'(x) = k\frac{e^{-kx}}{(1+e^{-kx})^2} = k\hat{v}(1-\hat{v})$$

$$\hat{v}''(x) = k^2\hat{v}(1-\hat{v})(1-2\hat{v})$$

hence (3) $\Leftrightarrow$

$$+ck\hat{v}(1-\hat{v}) = \nu k^2\hat{v}(1-\hat{v})(1-2\hat{v}) + b\,\hat{v}(1-\hat{v})(\hat{v}-a)$$

$$\Leftrightarrow \begin{cases} +ck = \nu k^2 - ba \\ 0 = -2\nu k^2 + b \end{cases} \Leftrightarrow \begin{cases} k = \pm\sqrt{\frac{b}{2\nu}} \\ c = \pm\sqrt{2\nu b}\left(\frac{1}{2} - a\right). \end{cases}$$

Hence $\hat{v}(x) = \frac{1}{1+e^{-\sqrt{\frac{b}{2\nu}}x}}$ is a tw-solution with wave speed and boundary conditions $\lim_{x\to-\infty} \hat{v}(x) = 0 = 1 - \lim_{x\to+\infty} \hat{v}(x)$.

<u>(Stochastic) stability of $\hat{v}$</u>

Suppose that $v(t,x) = u(t,x) + \hat{v}(x+ct)$ is a solution to (1). Then $u(t,x)$ satisfies the pde

$$(4) \qquad \partial_t u(t,x) = \nu \partial_{xx}^2 u(t,x) + b\left(f(u(t,x)+\hat{v}(x+ct)) - f(\hat{v}(x+ct))\right)$$

$$= \nu \partial_{xx}^2 u(t,x) + b f'(\hat{v}(x+ct))\, u(t,x) + R_b(x)$$

where the remainder $R_b(x) = \frac{b}{2} f''(\hat{v}(x+ct))\, u^2(t,x) + \frac{b}{6} f^{(3)}(\hat{v}(x+ct))\, u^3(t,x)$

$$= \frac{b}{2}\left(-6\hat{v}(x+ct) + 2(1+a)\right) u^2(t,x) - \frac{b}{6} a\, u^3(t,x)$$

satisfies

$$|R_b(x)| \le C\, u^2(t,x)\left(1+|u(t,x)|\right).$$

It is tempting to conjecture that $|u(0,x)| \ll 1$ then this implies $|u(t,x)| \ll 1$, i.e., $v(t,x)$ stays close to $\hat{v}(x+ct)$.

<u>Definition</u> The tw-solution $\hat{v}$ is called asymptotically stable in $L^2(\mathbb{R})$ if there exists $\delta > 0$ such that for $u(0,x) = v(0,x) - \hat{v}(x)$ in $L^2(\mathbb{R})$ with $\|v(0,\cdot) - \hat{v}\|_{L^2} \le \delta$ the unique variational solution $u(t,x)$ to (4) satisfies

$$\lim_{t\to\infty} \|v(t,\cdot) - \hat{v}(\cdot + c(t+t_0))\|_{L^2(\mathbb{R})} = 0$$

for some phase $t_0 \in \mathbb{R}$.

<u>Theorem:</u> $\exists\, \delta_0 > 0$ such that for initial condition $\|u(0,\cdot)\|_{L^2(\mathbb{R})} < \delta_0$

$$\|v(t,\cdot) - \hat{v}(\cdot + C(t) + ct)\|_{L^2(\mathbb{R})} \le e^{-\gamma_0 t} \|u(0,\cdot)\|_{L^2(\mathbb{R})}.$$

Here, $C(t)$ is the solution of the ode

$$(5) \quad C'(t) = -m \langle \partial_x \hat{v}(\cdot + C(t) + ct), \hat{v}(\cdot + C(t) + ct) - v(t,x) \rangle_{L^2(\mathbb{R}^d)}$$
$$C(0) = 0$$

for m sufficiently large.

Sketch of the proof: Take the variational solution of (4), then

$$\frac{d}{dt} \|u(t)\|^2_{L^2(\mathbb{R})} = 2\langle \nu \Delta u(t) + b f(u(t) + \hat{v}(t)) - f(\hat{v}(t)), u(t) \rangle$$

$$= 2\langle \nu \Delta u(t) + b f'(\hat{v}(t)) u(t), u(t) \rangle + \langle R_t(x), u(t) \rangle$$

$$\le b(4+a) \int |u(t,x)|^3 dx$$
$$\le b(4+a) \|u_x\|_{L^2(\mathbb{R})} \|u\|^2_{L^2}$$

Note that

$$\langle \nu \Delta u(t) + b f'(\hat{v}(t)) u(t), u(t) \rangle =$$

$$= \int -\nu (\partial_x u)^2(t,x) + b f'(\hat{v}(x+ct)) u^2(t,x)\, dx$$

and $f'(\hat{v}(x+ct)) < 0$ if $\hat{v}$ is close to 0 or 1, i.e. for $|x| \gg 1$.

(6) Provided $\langle \nu \Delta u(t) + b f'(\hat{v}(t)) u(t), u(t) \rangle \le -k_x \|u(t)\|^2_{H^{1,2}(\mathbb{R})}$

we could continue as

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|^2_{L^2(\mathbb{R})} \le -k_x \|u(t)\|^2_{H^{1,2}(\mathbb{R})} + b(4+a) \|u(t)\|_{H^{1,2}(\mathbb{R})} \|u(t)\|^2_{L^2(\mathbb{R})}$$

$$\le -\left(k_x - b(4+a) \|u(t)\|_{L^2(\mathbb{R})}\right) \|u(t)\|^2_{H^{1,2}(\mathbb{R})}.$$

Hence, provided $\|u(t)\|_{L^2(\mathbb{R})} \leq \frac{K_*}{b(4+a)}$, yields $\frac{d}{dt}\|u(t)\|^2_{L^2} < 0$,

hence the ball

$$B_{\frac{K_*}{b(4+a)}} = \left\{ u \in L^2(\mathbb{R}) : \|u\|_{L^2(\mathbb{R})} \leq \frac{K_*}{b(4+a)} \right\}$$

is <u>forward invariant</u> under the solution of the pde. and if $\delta < 1$ and

$$T := \inf \left\{ t \geq 0 : \|u(t)\|_{L^2(\mathbb{R})} \leq \delta \frac{K_*}{b(4+a)} \right\},$$

then $\frac{1}{2}\frac{d}{dt}\|u(t)\|^2_{L^2(\mathbb{R})} \leq -(1-\delta)\frac{K_*}{b(4+a)}\|u(t)\|^2_{H^{1,2}(\mathbb{R})}$

$$\leq -(1-\delta)\frac{K_*}{b(4+a)}\|u(t)\|^2_{L^2(\mathbb{R})}, \quad t \leq T$$

i.e.,

$$\|u(t)\|_{L^2(\mathbb{R})} \leq e^{-(1-\delta)\frac{K_*}{b(4+a)}t}\|u(0)\|_{L^2(\mathbb{R})}, \quad t \leq T,$$

and now T must be $+\infty$, hence exponential decay of $u(t)$ to zero.

However, (6) does not hold. Instead we have:

<u>Theorem:</u> let $K_* = \frac{2}{5}\frac{vb}{v+b}\, a(1-a)$ and $C_* = 6(v+b)$. Then

$$\langle v\Delta u + b f'(\hat{V})u, u\rangle \leq -K_*\|u\|^2_{H^{1,2}(\mathbb{R})} + C_*\langle u, \hat{V}_x\rangle^2.$$

<u>Interpretation:</u> $u(t)$ is not contracting in the direction of $\hat{V}_x$ which corresponds to the direction of motion of the tw (since $\frac{d}{dt}\hat{V}(x+ct) = c\hat{V}_x$)

We can think of the orbit

$$\mathcal{M} := \{\hat{v}(\cdot+\theta) : \theta \in \mathbb{R}\}$$

of the tw as a "manifold". Then $\hat{v}_x = \frac{d}{d\varepsilon}\hat{v}(x+\varepsilon)$ corresponds to the tangential direction and the "flow" generated by (4) is contracting in the "normal" direction only.

The last Thm now shows what to do: Introducing $c(t)$ yields

$$\frac{1}{2}\frac{d}{dt}\|\underbrace{v(t,\cdot)-\hat{v}(\cdot+c(t)+ct)}_{=:\tilde{u}(t)}\|^2_{L^2(\mathbb{R})} =$$

$$= \int -\nu(\partial_x\tilde{u}(t))^2 + b f'(\hat{v})\tilde{u}(t)^2\,dx$$

$$+ \langle \tilde{R}_{+,}\tilde{u}(t)\rangle + \underbrace{c'(t)\langle \hat{v}_x(\cdot+c(t)+ct), v(t,\cdot)-\hat{v}_x(\cdot}_{=-m\langle \hat{v}_x,\tilde{u}(t)\rangle^2}$$

$$\leq -K_*\|\tilde{u}(t)\|^2_{H^{1,2}(\mathbb{R})} + (C_*-m)\langle v_x,\tilde{u}(t)\rangle^2$$

$$+ b(4+a)\|\tilde{u}(t)\|_{H^{1,2}(\mathbb{R})}\|\tilde{u}(t)\|^2_{L^2(\mathbb{R})}$$

$$\leq -K_*\|\tilde{u}(t)\|^2_{H^{1,2}(\mathbb{R})} + b(4+a)\|\tilde{u}(t)\|_{H^{1,2}(\mathbb{R})}\|\tilde{u}(t)\|_{H}$$

and now one can proceed as before. ∎

Stochastic Stability

$$dv(t,x) = [\nu \partial_{xx} v(t,x) + b f(v(t,x))]dt + \varepsilon dW(t,x)$$

Then we can decompose

$$v(t,x) = \hat{v}(x + ct + C(t)) + u(t,x)$$

where C is as above the solution to

$$C'(t) = -m \langle \hat{v}_x(\cdot + C(t) + ct), \hat{v}(\cdot + C(t) + ct) - v(t,x) \rangle$$

and now $u(t,x) \approx \varepsilon Z(t,x)$, where

$$dZ_t = (\nu \partial_{xx}^2 + b f'(\hat{v}(\cdot + C(t) + ct)))Z_t dt + \varepsilon dW_t$$

is some (stable) Ornstein-Uhlenbeck process. Moreover, for

$$T = \inf\left\{ t \geq 0 : \|u_t\|_{L^2(\mathbb{R})} \vee \left(\int_0^t \|u_s\|^2_{H^{-1,2}(\mathbb{R})} ds \right)^{\frac{1}{2}} \geq \varepsilon^{-q} \right\}$$

then for $q < \frac{1}{2}$: $\sup_{t \leq T} \|u_t - \varepsilon Z_t\|_{L^2(\mathbb{R})} \leq C \varepsilon^{1-2q}$.